SUSE Linux Enterprise Server (SLES)

Offline Package Installation Guide: Installing Packages (e.g., `libcap`) from ISO/DVD Without an Active SUSE Subscription

1. Overview & Objective

In high-security, air-gapped, or trial enterprise environments, servers running **SUSE Linux Enterprise Server (SLES)** may not have access to the public internet or an active SUSE Customer Center (SCC) subscription. When trying to install packages using default repositories, the package manager (`zypper`) fails with registration errors.

This technical guide provides the exact procedure to utilize the **official SUSE installation ISO file or physical DVD** as a fully functional local offline repository to install packages such as **`libcap`**, **`libcap-progs`**, or any other core system library without requiring a commercial subscription or external internet connectivity.

> **Key Advantage of Using the Installation Media**
> The official SUSE ISO contains thousands of signed, official enterprise RPM packages. By mounting the media as a local repository, `zypper` automatically resolves internal dependencies, validates cryptographic hashes, and installs the exact versions supported by the operating system kernel.

2. Quick Command Reference (For `/mnt/dvd`)

If you have already mounted your SUSE installation ISO or DVD to `/mnt/dvd`, execute the following commands in sequence as root or with `sudo`:

Step / Action	Command to Execute	Purpose
1. Add Local Repository	`sudo zypper addrepo -c /mnt/dvd "SUSE_Local_ISO"`	Registers `/mnt/dvd` into Zypper package manager
2. Refresh Repositories	`sudo zypper refresh`	Reads package catalogs and metadata (press 'a' to trust)
3. Install libcap Packages	`sudo zypper install -y libcap2 libcap-progs`	Installs library, `setcap`, and `getcap` utilities
4. Verify Installation	`rpm -q libcap2 libcap-progs && which setcap`	Confirms package installation and executable paths

3. Detailed Step-by-Step Implementation

Step 3.1: Mount the ISO or DVD Media

If the ISO is a file on the disk (e.g. `/root/SLE-15-SP4-Full-x86_64.iso`), mount it as a loopback filesystem to `/mnt/dvd`:

```
# Create the mount point if it does not already exist
sudo mkdir -p /mnt/dvd

# Mount the ISO image (replace with your ISO file location)
sudo mount -o loop /path/to/SUSE_installation.iso /mnt/dvd

# Verify that the directory contains the media contents
ls -la /mnt/dvd
```

Step 3.2: Register the Local Repository in Zypper

Inform the Zypper package manager of the new repository on `/mnt/dvd`:

```
sudo zypper addrepo -c /mnt/dvd "SUSE_Local_ISO"
```

The `-c` (or `--check`) flag validates that a valid repository metadata catalog is located at that path.

Step 3.3: Refresh Zypper Metadata

Update the package database from the newly added local repository:

```
sudo zypper refresh
```

When prompted to trust the repository signing key:

```
New repository or package signing key received:
Do you want to (a)lways trust, trust (t)emporarily, or (d)isown the key? [a/t/d/?] (a): a
```

Type `a` (Always trust) and press Enter.

3. Detailed Step-by-Step Implementation (Continued)

Step 3.4: Search and Install `libcap`

Search for the available `libcap` packages in the local repository:

```
sudo zypper search libcap
```

Proceed to install the core library (`libcap2`) and userland tools (`libcap-progs`):

```
sudo zypper install -y libcap2 libcap-progs
```

Step 3.5: Verify Capability Utilities

Verify that the package is properly registered in the RPM database and verify binary utilities:

```
# Check RPM database
rpm -q libcap2 libcap-progs

# Confirm setcap and getcap tools are available
which setcap getcap

# Test capability assignment support
setcap -v cap_net_bind_service=+ep /path/to/binary || echo "Cap utilities active"
```

4. Alternative Method: Direct RPM Package Installation

If you prefer not to register a repository in Zypper, you can extract and install the exact RPM files directly from the mounted media:

```
# 1. Search for the exact libcap RPM files within the mounted ISO
find /mnt/dvd -name "*libcap*.rpm"

# Example output found on SLES installation media:
# /mnt/dvd/x86_64/libcap2-2.xx-xx.x86_64.rpm
# /mnt/dvd/x86_64/libcap-progs-2.xx-xx.x86_64.rpm

# 2. Install directly using zypper (resolves any local sub-dependencies)
sudo zypper install /mnt/dvd/x86_64/libcap2-*.rpm /mnt/dvd/x86_64/libcap-progs-*.rpm

# OR using standalone RPM tool directly:
sudo rpm -Uvh /mnt/dvd/x86_64/libcap2-*.rpm /mnt/dvd/x86_64/libcap-progs-*.rpm
```

5. Troubleshooting & Special Cases

Issue 1: "Repository type can not be determined"

On some versions of SLES (especially SLES 12/15 unified installer media), you may need to explicitly inform Zypper that the repository follows YaST2 media layout:

```
# Explicitly define YaST2 repository format
sudo zypper addrepo -t yast2 /mnt/dvd "SUSE_Local_ISO"
```

Issue 2: "Mount: /mnt/dvd is busy"

If you need to unmount the media after installation, make sure no terminals or processes are inside `/mnt/dvd` :

```
cd /root
sudo umount /mnt/dvd
```

Issue 3: Managing Existing Unregistered Repositories

If Zypper attempts to contact external SUSE servers that fail due to missing subscription, you can disable the external repositories temporarily or prioritize the local ISO:

```
# List all repositories and their status
zypper lr -d

# Set high priority (lower number = higher priority) for Local ISO
sudo zypper mr -p 50 "SUSE_Local_ISO"
```

6. Post-Installation Cleanup & Maintenance

Action	Command	Recommendation
Disable Local Repo	`sudo zypper mr -d "SUSE_Local_ISO"`	Disable when DVD is unmounted to prevent warning messages
Enable Local Repo	`sudo zypper mr -e "SUSE_Local_ISO"`	Re-enable anytime additional packages need to be installed
Remove Local Repo	`sudo zypper removerepo "SUSE_Local_ISO"`	Use if installation media is permanently removed from the server
Unmount Media	`sudo umount /mnt/dvd`	Frees the loop device and disc image

Enterprise Security & Compliance Verification

Installing packages from the official SUSE ISO preserves the integrity of your server's security baselines. Every package installed via this SOP has been cryptographically signed with the official SUSE GPG package signing key, ensuring zero risk of unvetted or compromised third-party binaries.